



# Quality Assurance Driven by Artificial Intelligence

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## Abstract

Customers are more digitally savvy than ever before. They expect more from technology, requiring it to become more intuitive and deliver greater benefits.

Artificial Intelligence (AI) in Quality Assurance is a game changer in today's SQA world. Machine Learning algorithms has helped in achieving greater standards in terms of enhanced Customer Focus, Cost Optimisation & Superior Customer Satisfaction.

This paper emphasizes specifics of leveraging Artificial Intelligence in Quality Assurance.

## Science behind AI QA Platform

*Human Intelligence + Machine Intelligence = AI QA platform.* Traditional QA has helped today's digital world gain a lot of wisdom. With this acquired knowledge and leveraging the power of Machine Learning Algorithms a superior quality assurance function is reborn.

*Correlation -> Mining Pattern -> Descriptive / Prescriptive / Predictive Insights -> Knowledge Base*

**Correlate Raw Data** – Inputs from business requirements, test cases, unit testing, defects logged, xRTM (extended RTM – mapping functionality to business objectives) and other test assets that may be available in an enterprise are fed into the big data platform using connectors. Natural Language Processing (NLP) technique is applied to interpret the raw data received from various sources. Machine learning algorithms correlate the data.

**Pattern Mining** – Machine learning algorithms self-learn and upskill the patterns to derive meaningful insights that is valuable for business sponsors and test engineers. Reinforcement Learning techniques are leveraged to mine the patterns from the raw data that are fed into the platform.

**Descriptive / Prescriptive / Predictive Insights** – Based on the correlated data & patterns mined, descriptive (What has happened), prescriptive (What should be done) and predictive (What could happen) insights are revealed for appropriate actions.

**Knowledge Base** – The success of any artificial intelligence platform can be measured by the volume of knowledge the platform has aggregated over a period of time. This is utilised to provide prescriptive and predictive insights.

## Benefits earned from AI QA Platform

The advantages of using an AI driven QA platform is huge. Key advantages that drive an SQA organisation to success are listed below.

**Reduce Test Fatigue** – Saving 15% to 20% of effort by removing duplicate test cases, definitely helps in improved productivity. Platform provides insight on the test cases that should be executed based on the risk identified using machine learning algorithms for the functionality under test.

**Extended Traceability** – Plotting criticality of business objectives and functionality against vulnerabilities helps in determining a Go / No go decision. Primarily integrated with Release Management function and provides higher customer satisfaction as the product is released to market with ZERO vulnerabilities.

**Business Process Assurance** – Platform assures that applications and services meet the demands of the business and customers by validating individual features and functions. The platform intelligently compares core business processes across the enterprise and plots application or services on a risk matrix visually. This unified dashboard provides a complete view of risk and vulnerabilities across various business processes within the enterprise.

**Predicting Weak Spots** – Predicting areas of failure and providing insights to engineers, on functionalities that need more test coverage. Platform learns from historical events that has been occurring for the application under test and provides insights.

### Values Attained

Metrics that can be derived from AI platforms are many. But there are perceptible results that accelerate the performance of a SQA organisation by leveraging AI platform.

**Customer Focus** – Insights projected are driven from a customer's eye.

**Cost Optimisation** – Near real time Predictive and Descriptive analytics drives productivity improvements.

**Superior Customer Satisfaction** – Zero vulnerabilities in business process and functionality drive better customer satisfaction.

### Conclusion: Leverage Data to Drive, Smarter & Faster Quality Assurance

Organisations demand high quality products and at a faster turnaround time. Artificial intelligence has entered our day-to-day living, and this can help organisations deliver high quality outputs with minimal or zero manual intervention.

Knowledge in various facets of quality assurance is key to build and implement an AI QA platform for any organisation. By partnering with GAVS, which has attained highest maturity in SQA functions, customers can efficiently embrace the power of Artificial Intelligence in SQA.

## About GAVS

GAVS Technologies (GAVS) is a global IT services & solutions provider enabling digital transformation through automation-led IT infrastructure solutions. Our offerings are powered by Smart Machines, DevOps & Predictive Analytics and aligned to improve user experience by 10X and reduce resource utilization by 40%.

For more information on how GAVS can help solve your business problems, write to [inquiry@gavstech.com](mailto:inquiry@gavstech.com)  
[www.gavstech.com](http://www.gavstech.com)

The logo for GAVS, featuring the letters 'GAVS' in a bold, white, sans-serif font against a dark blue background.